a — Introduction to WAN Technologies: WAN Connections, CSU/DSU, WAN Aggregators

Theory: Definition of WAN, types of WAN connections, WAN devices: CSU/DSU, WAN aggregators.

A WAN (Wide Area Network) connects separate LANs over large geographic distances (cities, countries) using links leased from service providers.

1. Comparison of WAN connection types

Type	Speed	Cost	Typical use cases
Leased line (e.g. T1/E1)	Fixed, dedicated: 1.5–2 Mbps per T1/E1, higher with fibre leased lines	High — you pay for the dedicated circuit whether used or not	Head office to data center, banks, links needing guaranteed bandwidth
DSL	Moderate: roughly 1–100 Mbps depending on DSL type and distance	Low	Small branch offices, backup links, home/remote workers
MPLS	High and scalable (Mbps to Gbps), with QoS	Medium to high	Multi-branch companies needing reliable, prioritised traffic (voice, video, ERP)

2. CSU/DSU connecting a router to a WAN service provider

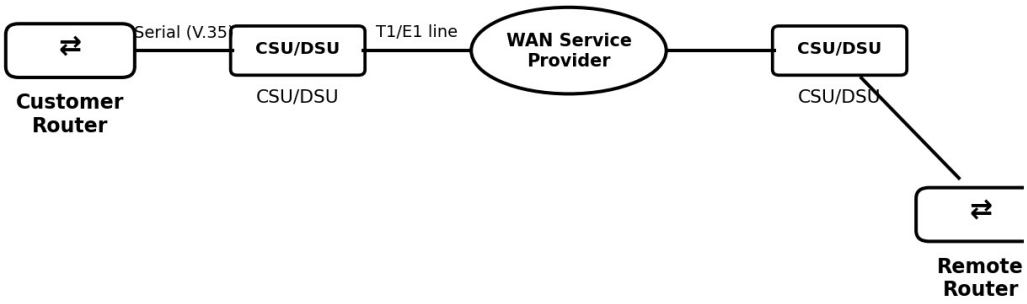


Fig: Router connected to a WAN service provider through a CSU/DSU (illustrative)

Router: the customer's router sends data from the LAN toward the WAN and chooses where packets go. CSU/DSU: it converts the router's serial (DTE) signals into the line format the provider's digital circuit (T1/E1) uses, provides clocking, and the CSU part protects and tests the line. WAN service provider: carries the traffic across its network to the remote site's CSU/DSU and router.

3. Three real-world apps that rely on WAN connections

- IRCTC: your booking request travels over the WAN/internet from your city to IRCTC's data centers, and payment gateways/bank networks are also reached over WAN links, so tickets can be booked from anywhere.
- WhatsApp: messages and media are sent over WAN links between your phone, WhatsApp's servers in remote data centers, and the receiver's phone, which is why chatting works across cities and countries.
- Zomato: the app in your hand, the restaurant's order screen and the delivery partner's app all connect to Zomato's cloud servers across WAN links, keeping orders and live tracking in sync.

4. WAN aggregator device (image/datasheet and key features)

Example device: Cisco ASR 1000 Series Aggregation Services Router (search this model name on Cisco's website to get its image and datasheet link).

- Key feature 1: high throughput and many WAN links - it aggregates traffic from many branch/WAN connections into one device, handling large volumes in hardware.
- Key feature 2: redundancy and scalability - supports redundant processors/power and modular interfaces, so a large enterprise or ISP edge can keep running and grow without replacing the device.

b — WAN Authentication Protocols (HDLC, PPP, PAP, CHAP, MS-CHAP)

Theory: Why authentication is needed in WAN. WAN authentication protocols: HDLC, PPP authentication (PAP, CHAP, MS-CHAP).

1. Network diagram: where authentication is needed

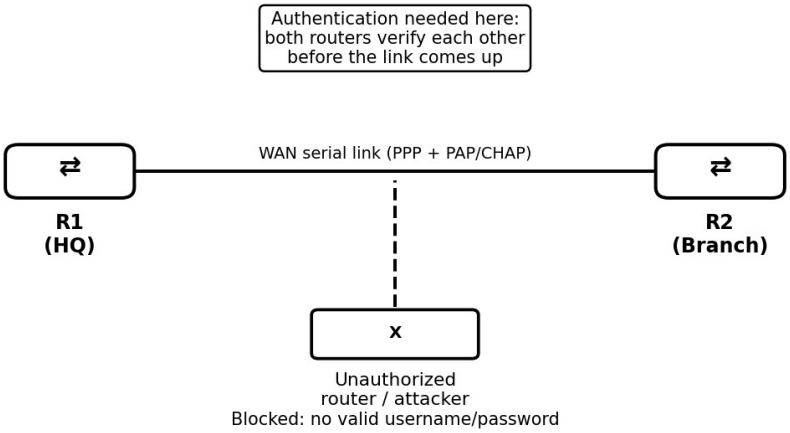


Fig: Authentication on a WAN link between two routers (illustrative)

Authentication is needed on the WAN link so each router proves who it is before the link is accepted; otherwise an unauthorized router or attacker could connect to the WAN link and receive or inject traffic.

2. HDLC WAN connection (and why it has no authentication)

```
R1(config)# interface serial 0/0/0
R1(config-if)# ip address 10.0.0.1 255.255.255.252
R1(config-if)# encapsulation hdlc
R1(config-if)# clock rate 64000     ! DCE side only
R1(config-if)# no shutdown

R2(config)# interface serial 0/0/0
R2(config-if)# ip address 10.0.0.2 255.255.255.252
R2(config-if)# encapsulation hdlc
R2(config-if)# no shutdown
```

The link comes up and pings work, but no username or password was ever asked for. HDLC (Cisco's version is proprietary) only frames the data; it has no authentication feature, so any router configured with HDLC and a matching address can join the link. Missing in terms of security: identity checking.

3. PPP with PAP authentication

```
R1(config)# username R2 password cisco
R1(config)# interface serial 0/0/0
R1(config-if)# encapsulation ppp
R1(config-if)# ppp authentication pap
R1(config-if)# ppp pap sent-username R1 password cisco

R2(config)# username R1 password cisco
R2(config)# interface serial 0/0/0
R2(config-if)# encapsulation ppp
R2(config-if)# ppp authentication pap
R2(config-if)# ppp pap sent-username R2 password cisco
```

If the password is incorrect on either side, PPP authentication fails: the interface stays 'up' but the line protocol goes 'down', pings fail, and debug ppp authentication shows a PAP authentication failure message. The link only comes up again once the passwords match.

4. Changing to CHAP

```
R1(config)# interface serial 0/0/0
R1(config-if)# ppp authentication chap

R2(config)# interface serial 0/0/0
R2(config-if)# ppp authentication chap

! username R2 / username R1 entries use the same shared password on both routers
```

Key difference: PAP sends the username and password in clear text, only once, when the link starts. CHAP never sends the password; the router sends a random challenge, the other side replies with a hashed (MD5) response, and the challenge is repeated periodically, so a captured packet cannot simply be replayed.

5. Real-world scenarios using these protocols

1. Mobile data/broadband login: when a phone or home router connects to an ISP over PPP/PPPoE, the ISP uses PAP/CHAP to check the subscriber's credentials. This is critical so only paying customers use the service and usage is billed to the right account.
2. Public Wi-Fi login (e.g. airport): the login page credentials are checked against an authentication server (often with CHAP/MS-CHAP style exchanges inside the Wi-Fi/RADIUS process). Authentication is critical so unknown people cannot join and misuse the network or the other users' traffic.

c — AAA (Authentication, Authorization, Accounting)

Theory: Three core functions, AAA in networking and security, AAA deployment models.

1. Scenario: user connecting to a private campus/airport Wi-Fi

Authentication: the user selects the Wi-Fi and enters a student ID and password. The access point passes them to the AAA server, which checks them against its database; wrong credentials = no access.
Authorization: once identity is proven, the server decides what the user may do — e.g. a student gets internet and library resources but not the admin network, and may be limited to certain sites or hours.
Accounting: while connected, the system records when the user logged in, how long they stayed, the device and the amount of data used. This log is used for security audits, troubleshooting and (at airports) usage limits or billing.

2. AAA deployment models (Zomato-type company)

	Centralized AAA (dedicated server, e.g. RADIUS/TACACS+)	**Local / distributed AAA (users stored on each device)**
Advantages	1. One place to add/remove users and change policy 2. Consistent rules and complete central logs for audits 3. Easily scales to many devices and offices	1. No extra server needed, simple for small setups 2. Works even if the network to the server fails 3. Low cost and quick to configure
Disadvantages	1. Server is a single point of failure unless made redundant 2. Needs extra hardware, setup and maintenance 3. Server holds sensitive data so becomes a target and must be heavily protected	1. User accounts must be changed manually on every device 2. Inconsistent policies and hard to audit 3. Does not scale; poor for a company holding sensitive customer data

3. Network diagram: AAA servers in an enterprise network

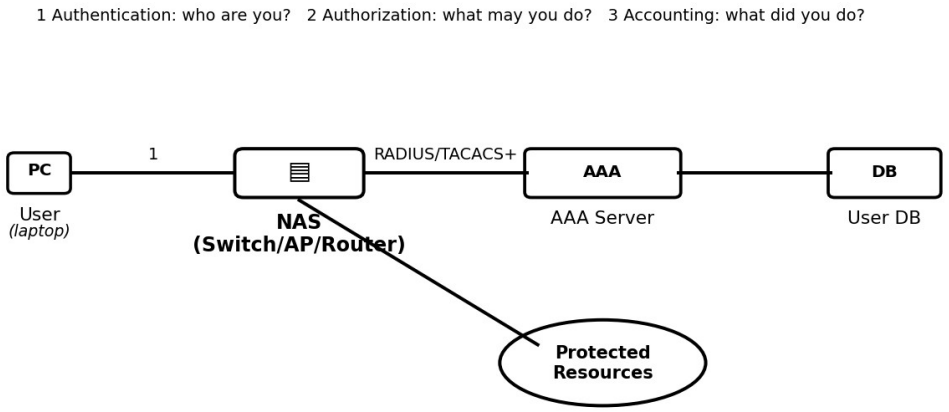


Fig: AAA server integrated into an enterprise network (illustrative)

Authentication happens when the user's credentials are sent from the NAS (switch/AP/router) to the AAA server and checked against the user database. Authorization happens next, when the server replies with what the user may access. Accounting happens throughout the session, as the NAS sends usage records to the server.

4. AAA applied to an app (Paytm)

When a user logs into Paytm, authentication confirms identity with the mobile number, password/PIN and OTP or biometric. Authorization then decides what that verified user can do — for example, a basic account may have lower transaction limits than a full-KYC account, and only the account holder can send money from their wallet. Accounting records every action: login time and device, each payment, and each failed attempt. These records support fraud detection, dispute resolution, statements and regulatory audits, much like the tracking used for security or billing.

5. Sample AAA policy for a college Wi-Fi (AI-generated sample)

This sample policy was drafted with an AI tool for the assignment; review it before submitting.

```
COLLEGE Wi-Fi AAA POLICY
1. Authentication: Only students with a valid college ID and password
   (verified against the college directory via a RADIUS server) may connect.
   Guests and unknown devices are denied.
2. Authorization: Students may access the internet and educational sites;
   social-media, streaming and gaming sites are blocked during class hours.
   Staff-only systems are not reachable from the student SSID.
3. Accounting: Every login, logout, device MAC, IP address and data usage is
   logged. Failed attempts are logged and reviewed weekly; logs kept 12 months.
```

Authentication is addressed by the ID/password check and denial of unknown users; authorization by the allowed/blocked website rules and restricted network areas; accounting by the logging of all access attempts and usage.

d — RADIUS & TACACS+

Theory: TACACS+ and RADIUS features, RADIUS authentication process, TACACS+ use cases, RADIUS vs. TACACS+.

1. RADIUS vs TACACS+ comparison

Feature	RADIUS	TACACS+
Protocol type	Open standard (IETF); uses UDP	Cisco-developed (now widely supported); uses TCP
Encryption	Encrypts only the password in the packet	Encrypts the entire packet body
Ports used	UDP 1812/1813 (older 1645/1646)	TCP 49
Command authorization	Limited; combines authentication and authorization, not per-command	Separates AAA functions; supports per-command authorization for admins
Typical use	Network access for users (Wi-Fi, VPN, dial-in)	Administrator access to network devices

2. RADIUS authentication sequence for a Wi-Fi login

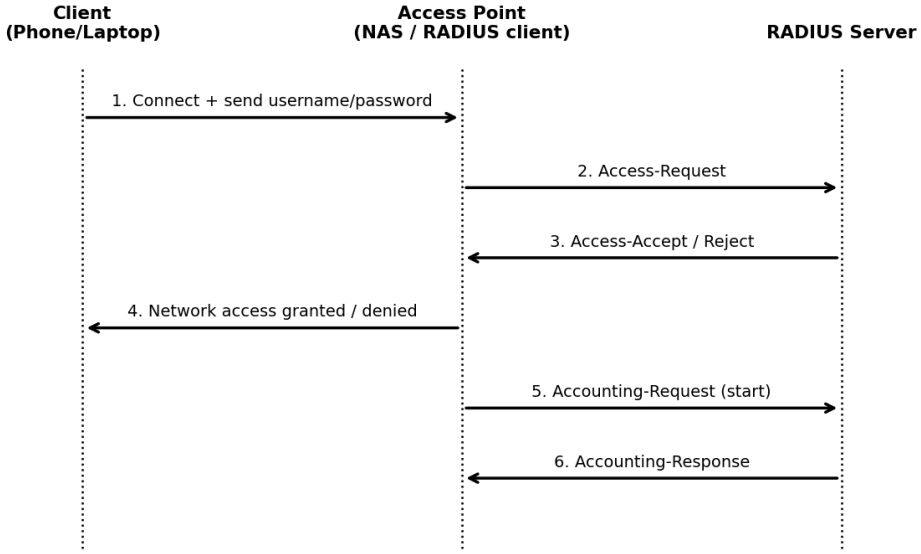


Fig: RADIUS authentication sequence for a Wi-Fi login (illustrative)

The user connects and enters credentials; the access point (RADIUS client) sends an Access-Request to the RADIUS server; the server checks the user database and returns Access-Accept or Access-Reject; the AP then grants or blocks the user and sends accounting messages while the session lasts.

3. Real-world scenario where TACACS+ is more suitable

In a company like IRCTC, the network team has many engineers logging in to routers and firewalls that carry booking traffic. TACACS+ is more suitable because it can authorize each command: a junior engineer may be allowed only 'show' commands while a senior engineer may change routing configuration. It also encrypts the whole session and logs every command an admin typed, so changes can be audited. RADIUS cannot control admin access with this level of detail.

4. RADIUS in a public hotspot (malls/airports)

Advantages:

- Centralized, scalable: one RADIUS server can authenticate users across many hotspot access points and locations.
- Open standard supported by almost all Wi-Fi equipment, with accounting data for usage limits and billing.

Disadvantages:

- Only the password is encrypted, so other information in the packet is exposed unless the link is protected (e.g. IPsec/RadSec).
- Uses UDP, which has no built-in delivery guarantee, and the server is a single point of failure unless made redundant.

5. Sample TACACS+ configuration for a Cisco router (AI-generated sample)

Sample generated with an AI tool using the prompt 'basic TACACS+ configuration for Cisco router'; lines to modify for a college lab are marked.

```
aaa new-model                                    ! enables AAA (needed first)
tacacs-server host 192.168.1.100 key LabKey123  ! CHANGE: lab server IP and key
aaa authentication login default group tacacs+ local
aaa authorization exec default group tacacs+ local
aaa accounting exec default start-stop group tacacs+
username labadmin privilege 15 secret LabAdmin!  ! CHANGE: local fallback account
line vty 0 4
  login authentication default
```

Lines to modify for a college lab: the tacacs-server host IP address and shared key must match the lab's TACACS+ server, and the local fallback username/password should be changed. The 'local' keyword is kept so staff can still log in if the server is unreachable. 'aaa new-model' stays as it is, since it turns on AAA.

e — Hybrid WAN (SD-WAN)

Theory: Introduction to SD-WAN, key benefits of SD-WAN.

1. Traditional WAN vs SD-WAN

Aspect	Traditional WAN	SD-WAN
How data is routed	Fixed paths, mostly over private links like MPLS, set by static routing	Dynamic path choice per application across MPLS, broadband and 4G/5G based on live link quality
Management	Each router configured separately at each site, often through CLI	Centralized controller/dashboard pushes policy to all sites
Security	Traffic usually backhauled to head office; security added by separate devices	Built-in encryption and segmentation, with security applied at each branch
Cost	Expensive private circuits and long provisioning times	Uses cheaper broadband alongside MPLS; faster, often zero-touch setup
Cloud access	Cloud traffic goes through the data center, adding delay	Direct local internet breakout to cloud apps like Office 365/AWS

2. SD-WAN diagram

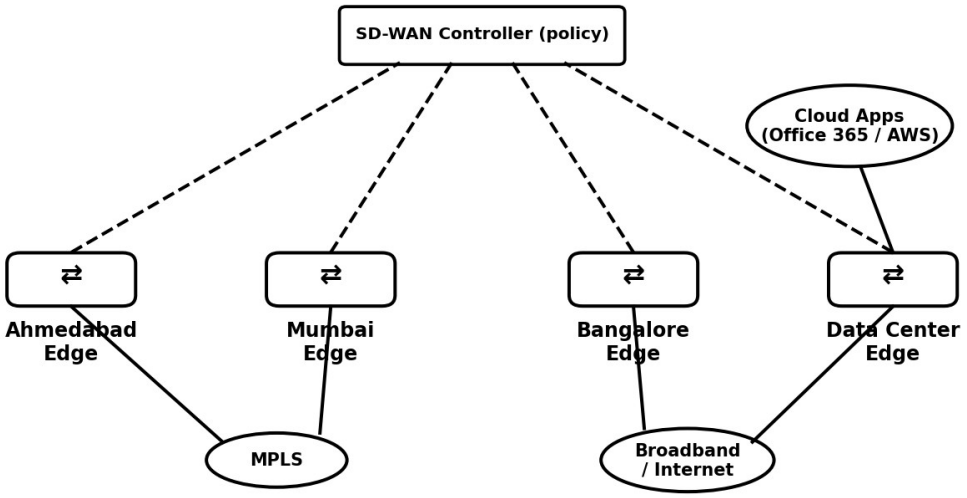


Fig: SD-WAN connecting branches, data center and cloud apps over MPLS and broadband (illustrative)

The SD-WAN controller (top) manages policy for all edge devices; each branch edge connects to the main data center and cloud apps over two WAN links: MPLS (reliable, private) and broadband/Internet (cheap, high bandwidth).

3. Scenario: Zomato across multiple cities

Imagine Zomato's offices and support centers in Ahmedabad, Mumbai and Bangalore all use Zomato's order and tracking systems at the same time. With SD-WAN, the order and live-tracking traffic is marked as high priority and automatically sent over the best-performing link, and if one link slows down or fails, traffic moves to the other within moments. Users and staff see faster, more reliable order updates, and security policies are applied the same way in every city.

4. Three key benefits of SD-WAN (Ahmedabad, Mumbai, Bangalore offices)

1. Lower cost: broadband can carry part of the traffic, reducing expensive MPLS usage. Example: the Ahmedabad office routes routine web and email over broadband and keeps MPLS for critical systems.
2. Better reliability and performance: it monitors links and moves traffic when one degrades. Example: if the Mumbai broadband link drops during a video call with Bangalore, the call shifts to MPLS or another link without disconnecting.

3. Simple central management and security: one dashboard applies policies to every site. Example: IT can set up a new Bangalore branch in a day by shipping a pre-configured edge device, without sending an engineer from Ahmedabad.

f — AAA (Authentication, Authorization, Accounting)

Theory: Three core functions, AAA in networking and security, AAA deployment models.

1. Comparison table: Authentication vs Authorization vs Accounting

Function	What it answers	Real-world examples
Authentication	Who are you?	1. Instagram: logging in with password plus a 2-factor code. 2. Paytm: logging in with a PIN, OTP or fingerprint. 3. Zomato: logging in with a mobile number and OTP.
Authorization	What are you allowed to do?	1. Instagram: you can edit only your own posts and profile; business accounts also get insights. 2. Paytm: transaction limits depend on whether your KYC is complete. 3. Zomato: a restaurant partner can edit only its own menu; a delivery partner sees only assigned orders.
Accounting	What did you do, and when?	1. Instagram: the list of devices/locations where your account is logged in. 2. Paytm: transaction history and statements. 3. Zomato: order history and delivery logs used for billing and disputes.

2. Network diagram: AAA securing a college campus Wi-Fi

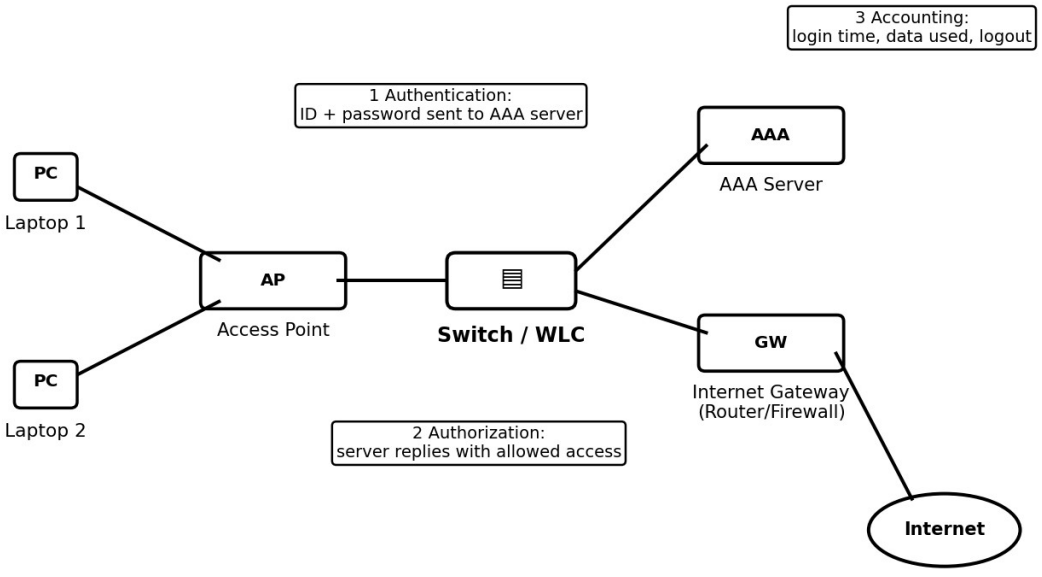


Fig: AAA securing access to a college campus Wi-Fi network (illustrative)

Authentication: the laptop's ID and password pass through the access point and switch to the AAA server, which checks them. Authorization: the AAA server replies with what the student may access (for example internet and library only). Accounting: the network device sends records of login time, data used and logout to the AAA server. Once allowed, the traffic goes through the internet gateway to the internet.

3. Centralized vs distributed AAA for multiple Flipkart warehouses

Centralized model: all warehouses send their login requests to one AAA server (or a redundant pair) at the head office. Staff accounts and rules are created once, so when an employee leaves, one change removes access to every warehouse, and all logs are in one place. If the link to the head office fails, logins at the warehouses can fail unless there is a backup.

Distributed model: each warehouse has its own AAA server (or stores users on its own devices). Logins keep working even if the head-office link is down, but accounts and policies must be managed separately at each warehouse, so they can become inconsistent, and the logs are spread across many places.

4. Centralized AAA for a large-scale app like BookMyShow

Advantages:

- One place to manage users and roles, so policy changes (for example, for staff at many theatres) are applied everywhere at once.
- Complete, central logs make it easier to detect fraud such as ticket-blocking abuse and to prepare audits.
- Scales well: new servers, offices or theatres simply point to the same AAA service.

Disadvantages:

- The server is a single point of failure during peak sales unless it is made redundant, which adds cost.
- It holds all credentials in one place, making it an attractive target that must be heavily secured.

5. Sample AAA policy for a Zomato-like app (AI-generated sample)

This sample policy was drafted with an AI tool for the assignment; the highlighted line is the one to change.

```
ZOMATO-LIKE APP: AAA POLICY
AUTHENTICATED ACTIONS
  - Customer: login (OTP), place order, change address, make payment
  - Restaurant partner: login (password), edit menu, view payouts
  - Delivery partner: login (OTP), accept order, update delivery status
  - Admin: login (password only)                    <<< CHANGE THIS
AUTHORIZED ROLES
  - Customer: own profile, own orders only
  - Restaurant partner: own restaurant menu and orders only
  - Delivery partner: only orders assigned to them
  - Admin: user management, refunds, system settings
ACCOUNTING LOGS
  - Logins (time, device, location), failed attempts
  - Orders and payments (who, what, amount, time)
  - All admin actions (who changed what, old and new value)
  - Logs retained for 12 months
```

One thing I would change to make it more secure: admins are authenticated with a password only. Admins can issue refunds and change system settings, so I would require multi-factor authentication (password plus an authenticator app or hardware key) for every admin login. A stolen password alone then cannot give an attacker admin access.

g — RADIUS & TACACS+

Theory: TACACS+ and RADIUS features, RADIUS authentication process, TACACS+ use cases, RADIUS vs. TACACS+.

1. Side-by-side feature comparison

Feature	RADIUS	TACACS+	Which protocol supports it
Open standard (IETF), works across most vendors	Yes	Originally Cisco-developed	RADIUS (TACACS+ is mainly Cisco-oriented, though others support it)
Encrypts the entire packet body	No - only the password	Yes	TACACS+
Per-command authorization for admins	Limited	Yes	TACACS+
Separates authentication, authorization and accounting	No - combines authentication and authorization	Yes - separate	TACACS+
Transport and ports	UDP 1812/1813	TCP 49	Both, with different transport
Network access (Wi-Fi 802.1X, VPN)	Yes - the standard choice	Not typical	RADIUS

2. Flow diagram: RADIUS authentication on a public Wi-Fi hotspot

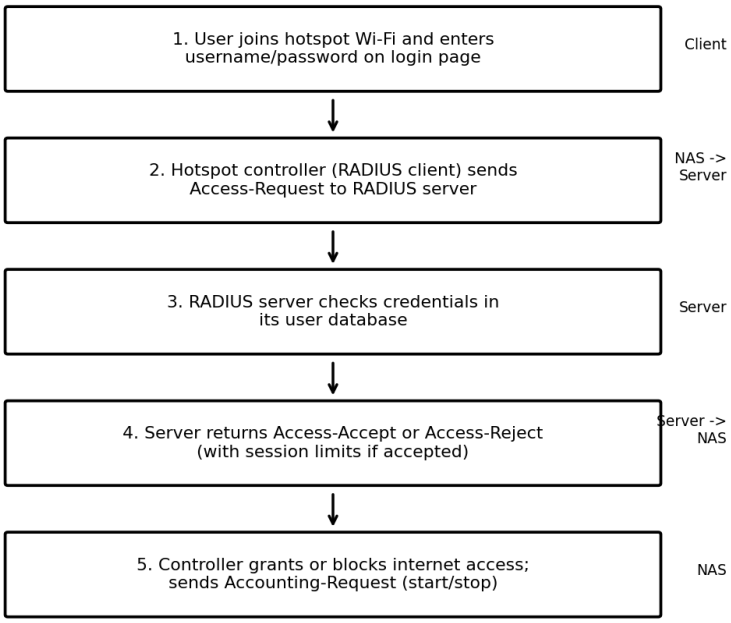


Fig: RADIUS authentication flow for a public Wi-Fi hotspot (illustrative)

Client to network device: the user enters credentials. Network device to RADIUS server: Access-Request. RADIUS server to network device: Access-Accept or Access-Reject. Network device to client: access granted or blocked, followed by accounting messages to the server.

3. Three use cases where TACACS+ is preferred over RADIUS

1. Administrator access to core routers and switches: TACACS+ can authorize each command, so a junior engineer is limited to 'show' commands while a senior engineer can change configuration. RADIUS cannot control commands at this level.

2. Large data centers with several teams: separate command sets can be given per role (network team, security team, auditors) from one server, because TACACS+ keeps authentication, authorization and accounting separate.
3. Banks and payment companies (like a Paytm-type environment) with strict audits: TACACS+ encrypts the whole session and records every command an admin typed, giving a full trail of who changed what.

4. A failed login to a network device: how each protocol responds

	RADIUS	**TACACS+**
Response	Server returns Access-Reject to the network device.	Server returns a FAIL/REJECT reply for the authentication step.
Logging of the failure	The failed attempt is recorded in the server's authentication log. Accounting normally covers sessions, so a failed login that never starts a session usually has no accounting record.	The failed attempt is logged with user, source device and time, in the same server that handles authorization and accounting.
Command authorization	Not applicable at this stage; RADIUS has no per-command control, so even after a successful login it cannot approve or deny individual commands.	No commands are authorized because login failed; after a successful login, every command could be authorized and its accounting record logged.

5. Sample RADIUS configuration for a Cisco router (AI-generated sample)

Sample generated with an AI tool; changes for a college campus Wi-Fi setup are listed after it.

```
aaa new-model
radius server COLLEGE-RADIUS
  address ipv4 192.168.1.100 auth-port 1812 acct-port 1813
  key LabKey123
aaa authentication login default group radius local
aaa authorization network default group radius
aaa accounting network default start-stop group radius
```

Changes I would make for a college campus Wi-Fi:

- Replace the server address with the campus RADIUS server, and use a long, random shared key instead of the sample key.
- Use 802.1X for Wi-Fi: add 'aaa authentication dot1x default group radius' and 'dot1x system-auth-control', and use WPA2-Enterprise on the SSID so each student logs in with their own ID.
- Add a second RADIUS server (backup) so students are not locked out if one server fails, and keep 'local' as a fallback only for the admin account.
- Keep accounting on so login times and data usage are logged for security reviews.